

Application of Parabolic Dish Collector in Automobile Industry

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Abstract

The automobile industry demands substantial process heat down the manufacturing line. This report presents a comprehensive study on integrating solar thermal energy into the automobile industry, specifically targeting the energy-intensive paint curing process. A Parabolic Dish Solar Collector (PDSC) system for meeting the thermal energy needs in paint curing was designed, analyzed, and compared with conventional heating systems. Detailed solar radiation modelling, thermal analysis and system sizing, performance analysis, and techno-economic evaluation were carried out. A 129 m² collector area with a 0.12 m² receiver was found optimal, achieving thermal outputs of 57 kW and operating temperatures of up to 270°C, using thermal oil (Therminol VP-1) as a heat transfer fluid. Economic and environmental analyses demonstrated annual savings of 522.88 GJ in energy, fuel savings of over 78% and CO₂ emission reductions of 31,260 kg annually. These results confirm the PDSC's potential for sustainable industrial heating applications.

Keywords: Solar Thermal Energy, Parabolic Dish Solar Collector (PDSC), Paint Curing, Industrial Process Heat, Renewable Energy, Energy Savings, CO₂ Emission Reduction, Automobile Industry, Thermal Efficiency

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1. Introduction

India produced 25.9 million vehicles in the year 2023, making it a substantial contributor to energy consumption in the industrial sector [1]. The paint shop plant in the automobile manufacturing process is a major consumer of heat, specifically, the paint curing process done in a curing oven. Paint curing can be done using hot air convection type (our case), radiation type, and electromagnetic wave curing type processes. Hot air curing involves passing a coated material through a continuous or batch oven where heated air at a temperature of approximately 190-220°C facilitates the curing of coatings, adhesives, or other applied materials and the energy required is traditionally supplied using electrical heaters or natural gas-fired systems [2].

In this study, we have designed a parabolic dish collector for the paint-curing process in the automobile industry, which reduces the use of electrical heaters. The challenge of ensuring a consistent supply of heat to plants while using all the solar energy available on the collector area can be addressed by the integration of solar thermal technology with existing heating sources, which is electrical heaters or natural gas-fired systems. Such an integration scheme utilizes full solar energy potential while being supported by conventional fuel that is used during low solar energy periods, nights and cloudy days. This project examines how solar thermal systems can be integrated to provide heating for the paint curing process [3]. This submission combines the work done previously and shows how solar thermal systems can be integrated to provide a heating paint curing process.

2. Application and Plant Configuration

Paint curing can be done using hot air convection type (our case), radiation type, and electromagnetic wave curing type processes. Hot air curing involves passing a coated material through a continuous or batch oven where heated air at a temperature of approximately 190–220°C facilitates the curing of coatings, adhesives, or other applied materials [2][4].

In this study, we have designed a parabolic dish collector for the paint curing process in an automobile industry, which reduces the use of electrical heaters. The annual energy required for the paint curing process comes out to 15.75 MWh, assuming operating hours in a day to be 14 hours (from 6:00 AM to 8:00 PM) and the plant is operational for 250 days in a year [2].

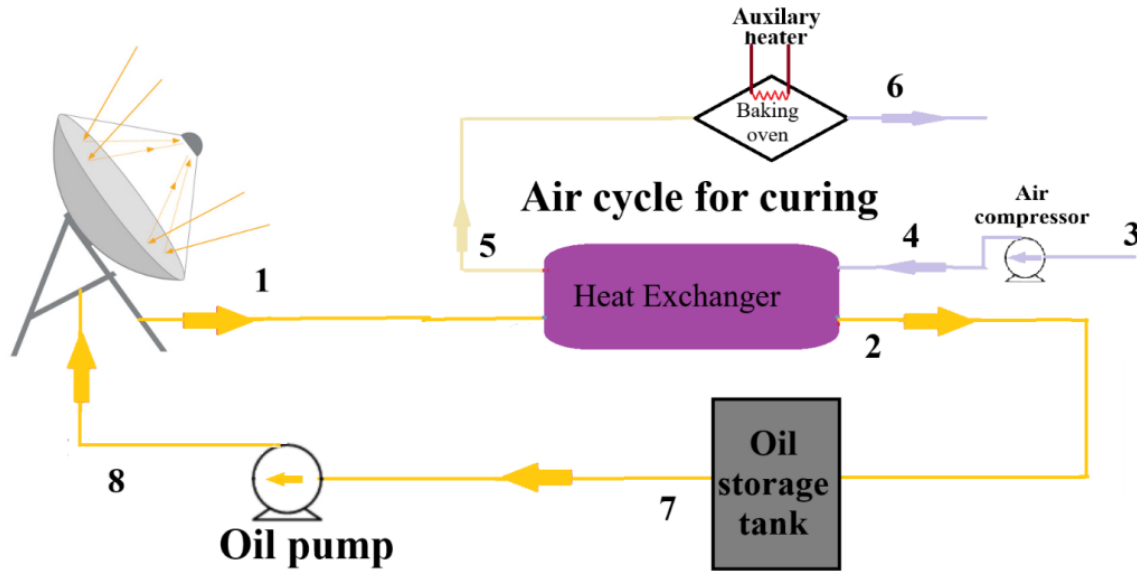


Figure 1: Process flow diagram of proposed solar thermal heating system

The parabolic dish collector consists of a parabola shaped collector which concentrates all the energy falling on it to a receiver which is kept at the focus of the receiver. The parabolic dish collector will be installed along with a 2D-tracking mechanism which will ensure that maximum incident beam radiation is falling on the collector. Parabolic dish collectors are usually used in high temperature heating applications.

Application type – Commercial

System Information:

Parabolic Dish Solar Collector

Thermal Oil medium or Heat Transfer fluid (HTF) – Therminol VP-1[5]

Operating Temperature ~ 200 °C - 290 °C [6]

Effectiveness for heat exchanger (ϵ) – 0.8

Parabola Dish Solar Collector consists of:

1. Collector Dish
2. Receiver
3. Dish and receiver support frame
4. Tracking mechanism
5. Piping and insulation
6. Pumps
7. Oil storage tank to control mass flow rate

Table 1: Table of required parameters (at design conditions)

Stream no.	Fluid	Mass Flow Rate (kg/s)	Temperature (°C)
1	Thermal oil	0.580	265

2	Thermal oil	0.580	220
4	Air	0.289	25
5	Air	0.289	220

3. Methodology

The radiation model is developed using the Liu-Jordan model and the data is obtained from NSRDB Data Viewer for Jodhpur. The obtained values of Direct Normal Irradiation (DNI) of Jodhpur are analyzed and plotted for better understanding.

Next, the processing plant conditions known from literature survey were used to calculate the area of Parabolic dish collector required to fulfil the heat requirements of paint curing for a design radiation value.

Finally, the annual energy analysis is done. Based on these results simple economic analysis was performed along with environmental impact calculation.

3.1 Radiation modelling

As our chosen solar collector is a dish-type solar collector with high concentration ratios, we can ignore the diffusion part of the solar energy. The design of our solar collector is based on the median DNI values for the months taken into consideration. The solar model used to calculate the DNI is as follows:

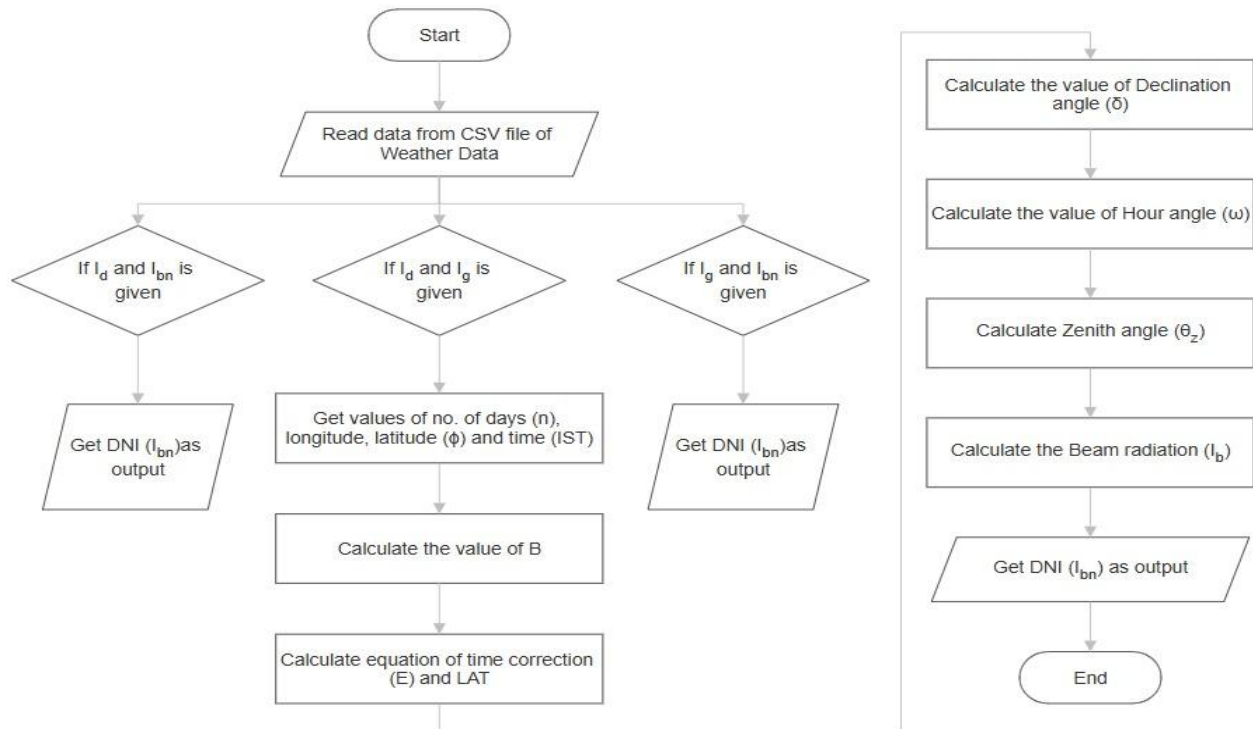


Figure 2: Flowchart of the Solar Radiation Model

Following are the equations used:

Local apparent time (LAT) or Solar time = Standard clock time – 4 × (Standard time longitude – longitude of location) + (Equation of time correction)

$$E = 229.18 \times [0.000075 + 0.001868 \cos(B) - 0.032077 \sin(B) - 0.014615 \cos(2B) - 0.04089 \sin(2B)] \quad (1)$$

Here,

$$B = (n - 1) \times \left[\frac{360}{365} \right] \quad (2)$$

n = day of the year

To get different angles used in solar geometry, the following equations are used:

$$\delta = 23.45 \times \sin \left[(284 + n) \left(\frac{360}{365} \right) \right] \quad (3)$$

Where,

n = Day of the year

$$\omega = (LAT - 12) \times 15^\circ \quad (4)$$

$$\cos \theta_z = \cos \delta \cos \omega \cos \phi + \sin \delta \sin \phi \quad (5)$$

$$I_{bn} = \frac{I_b}{\cos \theta_z} \quad (6)$$

Here, ϕ = Latitude of the location

According to the data for Jodhpur July and August are the lowest DNI producing months thus we removed those months while considering the design value of the system. Now we are considering the hours where we are getting enough DNI for the system to start generating heat, i.e., from 8 am to 4 pm for Jodhpur. We found the median of the value for DNI over the considered hours, except the months of August and July, which is 632.66 W/m^2 .

3.2 Plant modelling

Air flow rate calculation [6]:

Time required for each car to get cured in oven = 30 mins = 1800 secs

Mass of air required per body of car = 520 kg

Mass flow rate in the oven = $520 / 1800 = 0.289 \text{ kg/s}$

Heat requirement calculation:

$$\dot{Q} = \dot{m}_{air} \times \dot{C}_p \times (T_4 - T_3) \quad (7)$$

$$\dot{Q} = 56974.9049 \text{ W}$$

This is the heat transferred from oil to air in the heat exchanger. Now calculating the temperature inlet of oil required for the assumed effectiveness of the heat exchanger.

$$\epsilon = 0.8 = \frac{(\dot{m}C_p)_{air}(T_4 - T_3)}{(\dot{m}C_p)_{min}(T'_1 - T_3)} \quad (9)$$

Taking $(\dot{m}C_p)_{min} = (\dot{m}C_p)_{air} \Rightarrow T'_1 = 268.75 \text{ }^\circ\text{C}$

Approximating the $C_{p,oil,avg}$ to be the average of specific heat capacities [7] at temperature $220 \text{ }^\circ\text{C}$ to $290 \text{ }^\circ\text{C}$ as this range is a superset of the working temperature. Next, we solve for mass flow rate of thermal oil using heat balance across the two streams of heat exchanger.

$$\begin{aligned} (\dot{m}C_p)_{air}(T_4 - T_3) &= (\dot{m}C_p)_{oil}(T_1 - T_2) \\ \Rightarrow \dot{m}_{oil} &= 0.8978 \text{ kg/sec} \\ \text{Using } C_{p,oil,avg} &= 2207.28 \text{ J/kg-K} \end{aligned} \quad (10)$$

Now that we have the temperature of oil required for the needed amount of heat transfer and its mass flow rate, we assume here that 5% of heat is lost in flow from receiver to the heat exchanger inlet from the oil.

$$\begin{aligned} T_1 &= \frac{\dot{Q}}{(1-0.05) \times \dot{m}_{oil} \times C_{p,oil,avg}} + T_2 \\ T_1 &= 270.26 \text{ }^\circ\text{C} \end{aligned} \quad (11)$$

The losses from the receiver are the conduction, convection and the radiative losses. Generally, the conductive losses from the receiver are neglected because they are covered by insulation [8].

Now, for calculating the collector area and the concentration ratio of the plant, we have assumed the receiver area so as to have very less convective and radiative losses.

The assumed Concentration ratio (C) = 1100 [9].

$$Q_u = 56974.90 \quad (12)$$

Optical efficiency:

$$\eta_{opt} = \rho_{col} \cdot \alpha = 0.76912 \quad (13)$$

Average Receiver surface temperature:

$$T_{s,avg} = \frac{T_1 + T_8}{2} + 50 \quad (14)$$

Receiver area:

$$A_r = \frac{Q_u}{\eta_{opt} \cdot C \cdot I_d - \left(\epsilon_{rec} \cdot \sigma \cdot \left((T_{s,avg})^4 - (T_3)^4 \right) + h \cdot (T_{s,avg} - T_3) \right)} = 0.117 \text{ m}^2 \quad (15)$$

Collector area:

$$A_c = C \cdot A_r \quad (16)$$

Absorbed heat:

$$Q_{abs} = \eta_{opt} \cdot I_d \cdot A_c \quad (17)$$

Heat received by receiver:

$$Q_{rec} = \frac{Q_{abs}}{\alpha_{rec}} \quad (18)$$

Heat loss:

$$Q_{loss} = Q_{abs} - Q_u \quad (19)$$

Thermal efficiency of receiver:

$$\eta_{th_{rec}} = \frac{Q_u}{Q_{rec}} \quad (20)$$

Overall efficiency:

$$\eta_{overall} = \eta_{opt} \cdot \eta_{th_{rec}} \quad (21)$$

For calculating the heat loss from the receiver, a temperature increase of 50 °C on the receiver surface from the fluid temperature is taken [9].

The values taken are [10];

$$\epsilon_{rec} = 0.11$$

$$\alpha = 0.836$$

$$\rho_{col} = 0.92$$

$$I = 632.66 \text{ W/m}^2.$$

$$h = 25 \text{ W/m}^2 \text{ K}.$$

Thermal efficiency of the system, $\eta_{th} = \eta_{opt} \cdot \eta_{th,rec} = 76.19 \%$.

3.3 Part Load Analysis

The performance analysis of our designed collector system is tested by simulating its resulting values of output temperature of oil (T_2), output temperature of air (T_5), and required value of mass flow rate of oil. In our design for of load conditions where DNI is less than the designed DNI value the energy extracted will be less than designed value, thus we need to maintain the mass flow rate of oil such that the temperature of oil inlet in the HX is remained at same designed value that is 538 K (265°C). By changing the value of mass flow rate both the outlet streams have different values of temperature than designed. Temperature inlet of air in HX is taken to be 298K (25°C).

The solving(operation) algorithm works as follow:

1. If the DNI value at the given hour is less than 260.93 W/m². For the pointed values of DNI than 260.93 W/m², the code sees it as anomaly (this is due to the oil mass flow rate becoming too small and/or T_2 becoming less than T_5). Filtering out these data means for all those values the conventional heater is operated.

2. If the DNI value is more than the design value that is 631 W/m^2 . The tracker receives a signal to misalign the collector so that the focal region area overlapping the receiver decrease thus, energy falling on the receiver becomes equal to the designed value.
3. For other values of DNI the algorithm solves the values of outlet temperatures of both the streams, mass flow rate of oil to be maintained, heat exchanged in the HX, Q_{loss} in the receiver due to convective and radiative methods. As the energy given to the air will be less than the required value hence the temperature outlet of air from the heat exchanger will be less than needed. In this case the auxiliary heater turns on and heat the air to the required value that is 220°C . Thus, we calculate the required Q_{aux} value.

$$T_5 = \frac{Q_u}{\dot{m}_{\text{air}} \times C_{p,\text{air}}} + T_4 \quad (22)$$

$$T_2 = T_1 - \frac{Q_u}{\dot{m}_{\text{oil}} \times C_{p,\text{oil}}} \quad (23)$$

$$Q_{\text{loss}} = A_r \left(\epsilon \sigma \left[\left(\frac{T_1 + T_2}{2} + 20 \right)^4 - T_{\text{amb}}^4 \right] + h_{\text{air}} \left(\frac{T_1 + T_2}{2} + 20 - T_{\text{amb}} \right) \right) \quad (24)$$

$$Q_u = Q_{\text{abs}} - Q_{\text{loss}} \quad (25)$$

Here Q_u is the radiation absorbed by the receiver

$$\frac{UA}{UA_d} = \left(\frac{\dot{m}_{\text{oil}}}{\dot{m}_{\text{oil}_d}} \right)^{0.7} \quad (26)$$

Where, UA_d is found out using the below equation,

$$Q_{u_d} = UA_d \times LMTD_d \quad (27)$$

Here Q_{u_d} is equal to the load that is 57 kW and $LMTD_d$ is solved by

$$LMTD_d = \frac{(T_1 - T_{5_d}) - (T_2 - T_4)}{\ln \left(\frac{T_1 - T_{5_d}}{T_2 - T_4} \right)} \quad (28)$$

$$\dot{m}_{\text{oil}} \times C_{p,\text{oil}} \times (T_1 - T_2) = UA \times LMTD \quad (29)$$

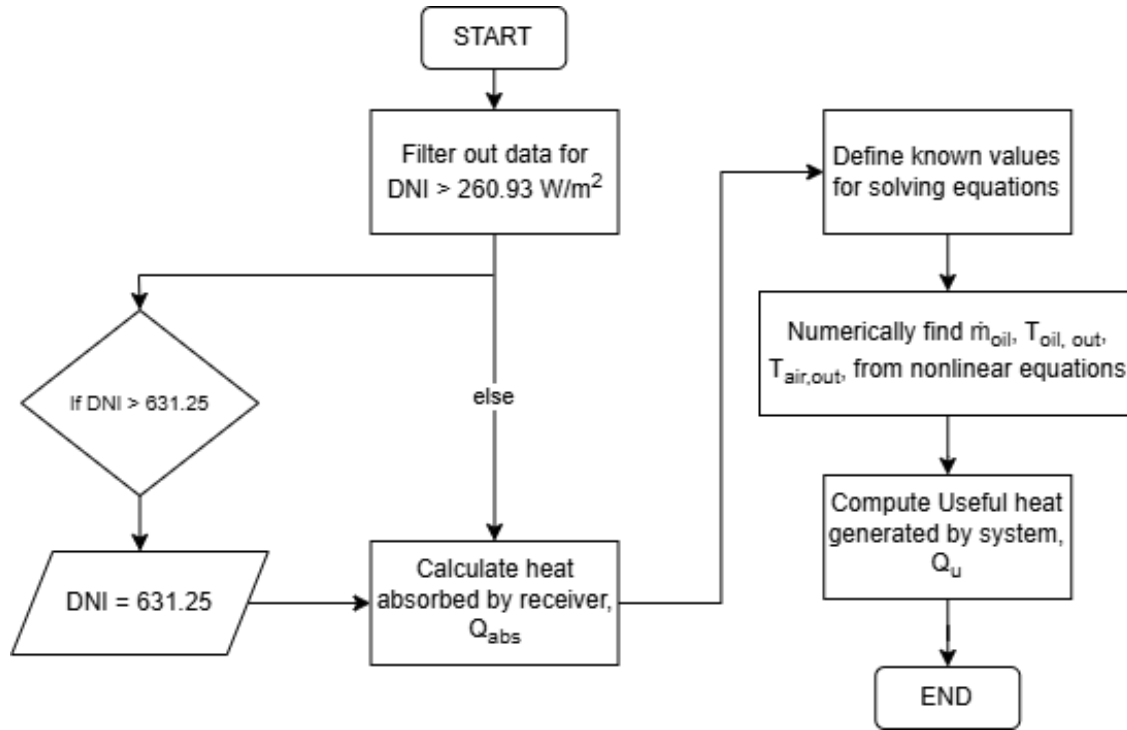


Figure 3: Flowchart for calculating part load operating parameters

3.4 Economic Analysis

We are replacing a convention natural gas fired boiler oven with a parabolic dish along with auxiliary conventional natural gas boiler.

Table 2: Parameters used for economic analysis

S. No	Parameter	Value	Unit
1	Collector	30100 [11]	Rs/m ²
2	Receiver	2679.76 [12]	Rs
3	Burner efficiency	0.9	-
4	Calorific Value	46	MJ/kg
5	Operating days	365	days
6	Operating hours	8	hrs/day
7	Rate of Natural Gas	344 [13]	Rs/MMBTU
8	Density of Natural Gas	0.76 [14]	Kg/SCM

Case-1: Without PDSC system

Power required to heat air = 56.9749 kW

Energy required per day = 56.9749 * operating hours = 455.8 kWh

Energy supplied by boiler = $\frac{\text{Energy required per day}}{\text{Boiler efficiency}}$ = 506.44 kWh

Annual energy supplied = Energy supplied by boiler * operating days * 3.6 * 10⁶ = 6.654 x 10¹¹ J

Mass of Natural Gas required = $\frac{\text{Annual Energy Supplied}}{\text{Calorific value}}$ = 14466.67 kg

1 MMBTU = 25.2 SCM [10]

$$\text{Cost of fuel required} = \frac{\text{Annual Energy Supplied}}{\text{Density of Natural Gas} * 25.2} * 344 = \text{Rs } 259844.11$$

Case-2: With PDSC

Annual energy supplied to heat air from PDSC was obtained from the performance model = $4.706 * 10^{11}$ J

Annual energy required to heat air from conventional boiler was obtained from the performance model = $1.283 * 10^{11}$ J

$$\text{Annual energy supplied by natural gas boiler} = \frac{1.283 \times 10^{11}}{0.9} = 1.425 * 10^{11} \text{ J}$$

$$\text{Mass of Natural Gas required} = \frac{\text{Annual Energy Supplied by boiler}}{\text{Calorific value}} = 3099.45 \text{ kg}$$

$$\text{Cost of fuel required} = \frac{\text{Annual Energy supplied by boiler}}{\text{Density of Natural Gas} * 25.2} * 344 = \text{Rs } 55671.13$$

$$\text{Energy saved from using PDSC} = \frac{\text{Annual energy supplied by PDSC}}{0.9} = 522.88 \text{ GJ}$$

$$\text{Energy savings from fuel} = \frac{\text{Energy saved from using PDSC}}{\text{Annual Energy supplied by boiler from case 1}} * 100 = 78.58\%$$

Table 3: Results from economic analysis

S. No	Parameter	Without PDSC	With PDSC	Savings
1	Energy supplied by boiler	665.4 GJ	142.55 GJ	522.88 GJ
2	Fuel	14466.67 kg	3099.45 kg	11,367.22 kg
3	Fuel Cost	259844.11 Rs	55671.13 Rs	2,04,173 Rs

PDSC capital cost = Area of collector * Collector cost + Area of receiver * Receiver cost

From above equation, we get capital cost of PDSC to be Rs 38,83,051.36

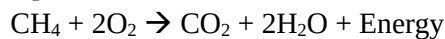
Assuming Other costs (such as tracking system, support chassis, etc) to be 10% of the capital cost will be Rs 3,88,305.13

$$SPP = \frac{\text{Capital cost} + \text{other costs}}{\text{Fuel cost savings}} \quad (30)$$

We get an SPP of 20.92 years after substituting the values in the above equation.

3.5 Environmental Analysis

Assuming complete combustion takes place in a boiler,



1 kg of Methane gives 2.75 kg of CO₂

Without PDSC, emission will be 39,783.34 kg of CO₂

With PDSC, emission will be 8,523.48 kg of CO₂

Emissions saved would be 31259.86 kg of CO₂

4. Results and Discussion

4.1 Thermal, Economic and Environmental Performance

The performance simulation was conducted for a full calendar year using hourly DNI data for Jodhpur. The simulation tracks variations in the useful energy output (Q_u), outlet temperatures of the HTF (T_2), and process air (T_5), as well as the activation of the auxiliary heater during off-design conditions.

Table 4: Design conditions

S. No	Parameter	Value	Unit
1	Design value of Irradiance	631.25	W/m^2
2	Emissivity of receiver	0.11	-
3	Max Effectiveness	0.8	-
4	Reflectivity of collector	0.85	-
5	Absorptivity	0.836	-
6	Concentration ratio	1100	-
7	*Tank capacity	2.287	m^3

*Tank capacity corresponds to working fluid supplied for 1 hr for maximum mass flow rate (that is 0.58)

Table 5: System specifications at design conditions

S. No	Parameter	Value	Unit
1	Thermal Power	56.975	kW
2	Thermal Efficiency	0.58	-
3	Dish Aperture Area	128.91	m^2
4	Receiver Aperture Area	0.1172	m^2
5	Mass flow rate of HTF	0.58	kg/s
6	Working temperature of HTF	265	$^{\circ}C$

The collector operates efficiently in the DNI range of 260.93 W/m^2 to 872.09 W/m^2 . During periods of low DNI (early mornings, late afternoons, and winter months), the auxiliary heater compensates for the shortfall in useful heat output. The oil mass flow rate dynamically adjusts to maintain the HTF outlet temperature at the designed 265 $^{\circ}C$. The corresponding air temperature after the heat exchanger ranges from 104.66 $^{\circ}C$ to 220 $^{\circ}C$.

The Q_u from the PDSC system is calculated to be approximately 4.706×10^{11} J. On a daily basis, this equates to around 1.26 MWh of thermal energy over an 8-hour operating window. The thermal efficiency of the receiver system was estimated to be 76.19%, while the overall system efficiency (including optical and heat exchanger losses) averaged at 58%.

From an economic standpoint, we compared two scenarios:

1. Without PDSC system: The entire thermal energy requirement is met using a conventional natural gas-fired burner. This results in a fuel consumption of 14,466.67 kg/year and a corresponding cost of ₹2,59,844.11 (at ₹344/MMBTU).
2. With PDSC system: The solar system supplies 78.58% of the required thermal energy, reducing the natural gas consumption to 3,099.45 kg/year, with a cost of ₹55,671.13.

This yields an annual fuel cost saving of approximately ₹2,04,173. Given a total capital investment of ₹38,83,051 (collector cost + receiver + 10% other components), the calculated Simple Payback Period (SPP) is 20.92 years. While this is relatively long, it is acceptable for systems with a 25+ year operational life and low maintenance.

Environmental Impact: Assuming methane (CH_4) as the primary component of natural gas, with 1 kg CH_4 producing 2.75 kg of CO_2 upon combustion, the carbon emissions are:

1. Without PDSC: 14,466.67 kg $\text{CH}_4 \rightarrow 39,783.34$ kg CO_2
2. With PDSC: 3,099.45 kg $\text{CH}_4 \rightarrow 8,523.48$ kg CO_2

Hence, the solar thermal system results in a reduction of 31,260 kg of CO_2 emissions annually. This highlights the potential of solar thermal energy in decarbonizing industrial processes.

4.2 Comparison with Literature and Commercial Systems

Table 6: Comparison with Solar dish concentrator systems of similar thermal loads

Model Name	Solarbeam 9M [9]	ARUN 160 [15]	Our system
Thermal Power	45 kW	~75 kW	~60 kW
Operating Temperature of working fluid	Upto 270 °C	Upto 300 °C	Upto 290 °C
Collector Area	63 m ²	169 m ²	129 m ²
Receiver Area	0.07 m ²	-----	0.117 m ²
Mass flow rate of working fluid	0.47 kg/s	0.83 kg/s	0.58 kg/s

Now based on collector area, our system's 129 m² collector area is significantly larger than SOLARBEAM 9M (63 m²), which produces only 45 kW. It is smaller than ARUN 160 (169 m², ~75 kW), but still within a comparable range considering its 57-kW output. This suggests that our system's efficiency is relatively high compared to SOLARBEAM 9M and approaches that of ARUN 160. Based on receiver area our receiver area of ~ 0.12 m² is comparable to SOLARBEAM 9M (0.07 m²), indicating a similar concentration ratio and heat transfer characteristics. For ARUN 160, receiver area is not specified, but given its larger collector area, the receiver could be larger or use a different design for heat absorption.

Our system performs competitively. It delivers higher thermal output than SOLARBEAM 9M while requiring less area than ARUN 160, indicating an optimized thermal design. The high concentration ratio (1100) and high receiver absorptivity contribute to its effective performance. Additionally, our system incorporates a 2D sun tracking system and dynamic control for part-load operation, which are typically missing in commercial off-the-shelf units.

The monthly average efficiency remains consistent due to the filtering of low-DNI months. Figures 4 and 5 illustrate the variation in monthly useful energy and efficiency, validating the robustness of the system across seasonal conditions.

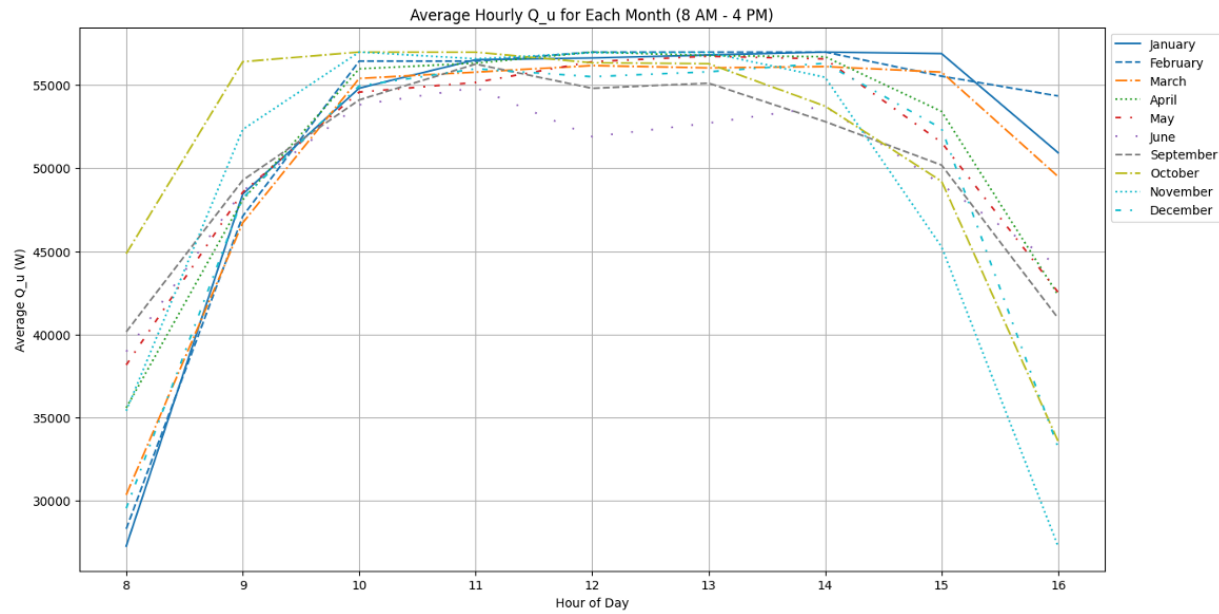


Figure 4: Variation in monthly useful energy hour wise

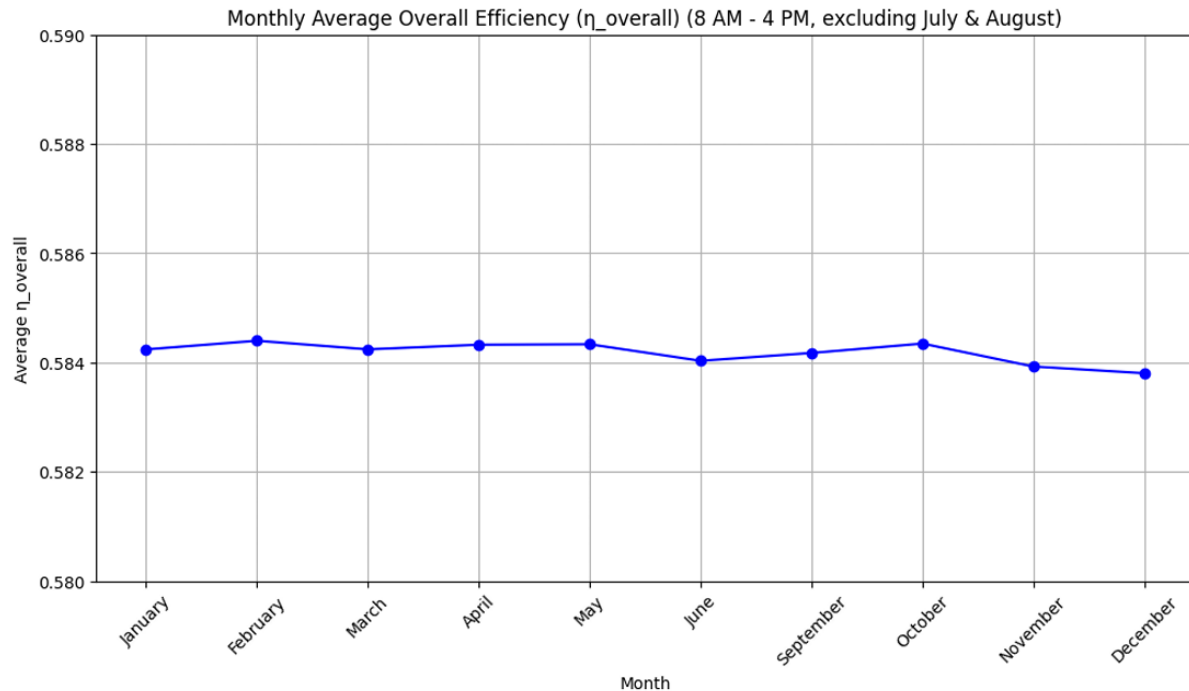


Figure 5: Variation of efficiency month wise

Overall, our analysis demonstrates that a well-designed PDSC system, operating under realistic weather and plant conditions, can significantly offset fossil fuel usage, reduce emissions, and deliver significant savings. This validates the potential of such systems for fulfilling heat requirements for various industrial processes.

Conclusion

Through this detailed study, we have demonstrated the feasibility, effectiveness, and practicality of deploying a PDSC system for meeting the thermal energy requirements of the paint curing process in the automobile industry.

The designed system, optimized for a plant located in Jodhpur, uses Therminol VP-1 as the working fluid and operates in the temperature range of 200–220°C. The collector system, with a 129 m² aperture area and a 0.12 m² cavity receiver, can generate up to 57 kW of thermal power while achieving 58% overall efficiency. Our radiation model, based on the Liu and Jordan formulation, enabled accurate simulation of hourly DNI values, which were further refined by excluding low-solar-output monsoon months.

The annual simulation results confirm that the PDSC system can supply approximately 4.706×10^{11} J of useful thermal energy, corresponding to nearly 78.6% of the plant's annual heating load. This significantly reduces the consumption of natural gas, resulting in yearly fuel cost savings of ₹2.04 lakhs. Although the payback period of approximately 20.92 years may appear long, it is justified by the low maintenance, long lifespan, and high reliability of the system.

From an environmental perspective, the reduction of over 31,260 kg of CO₂ annually emphasizes the role of such systems in promoting sustainable industrial operations. Furthermore, our system's performance compared favorably with established commercial alternatives like the SOLARBEAM 9M and ARUN 160, indicating a well-optimized balance between size, efficiency, and cost.

In conclusion, this study validates the role of high-temperature concentrating solar thermal systems in replacing fossil-fuel-based heating processes within Indian industries. With further enhancements such as thermal energy storage integration and multi-process coupling (e.g., combining curing with pre-treatment tanks), the system can be made more versatile and efficient.

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